

Question 1: Given a relation EMPLOYEE(EmpID, Name, Dept, Salary), write a relational algebra expression to retrieve all employees in the 'HR' department with Salary > 50000.

Aim: To retrieve the details of employees working in the HR department whose salary is greater than 50000 using relational algebra.

Relational Algebra Query

$$\sigma_{Dept='HR' \wedge Salary > 50000}(EMPLOYEE)$$

```
MySQL 8.0 Command Line Cli x + v

mysql> CREATE DATABASE CompanyDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE EMPLOYEE
-> (
->     EmpID INT PRIMARY KEY,
->     Name VARCHAR(50),
->     Dept VARCHAR(30),
->     Salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO EMPLOYEE VALUES
-> (101,'Rahul','HR',60000),
-> (102,'Priya','IT',75000),
-> (103,'Arun','HR',48000),
-> (104,'Meena','Finance',70000),
-> (105,'Kiran','HR',55000);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM EMPLOYEE;
+-----+
| EmpID | Name  | Dept  | Salary |
+-----+
| 101   | Rahul | HR    | 60000.00 |
| 102   | Priya | IT    | 75000.00 |
| 103   | Arun  | HR    | 48000.00 |
| 104   | Meena | Finance | 70000.00 |
| 105   | Kiran | HR    | 55000.00 |
+-----+
5 rows in set (0.00 sec)

mysql> SELECT *
-> FROM EMPLOYEE
-> WHERE Dept='HR' AND Salary>50000;
+-----+
| EmpID | Name  | Dept  | Salary |
+-----+
| 101   | Rahul | HR    | 60000.00 |
| 105   | Kiran | HR    | 55000.00 |
+-----+
2 rows in set (0.00 sec)
```

Question 2: Write SQL queries to create the STUDENT(SID, Name, Branch, CGPA) table, insert five records, and retrieve students with CGPA > 8.0 ordered by CGPA in descending order.

Aim: To create the STUDENT table, insert five records, and retrieve students whose CGPA is greater than 8.0 in descending order.

Create Table

CREATE TABLE STUDENT

(

SID INT PRIMARY KEY,

Name VARCHAR(50),

Branch VARCHAR(30),

CGPA DECIMAL(3,2)

);

Insert Records

INSERT INTO STUDENT VALUES

(101,'Rahul','CSE',8.5),

(102,'Priya','IT',9.1),

(103,'Arun','ECE',7.8),

(104,'Meena','AIDS',8.9),

(105,'Kiran','CSE',8.2);

Retrieve Students

SELECT *

FROM STUDENT

WHERE CGPA > 8.0

ORDER BY CGPA DESC;

```
MySQL 8.0 Command Line Cli x + v
mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE STUDENT
-> (
->     SID INT PRIMARY KEY,
->     Name VARCHAR(50),
->     Branch VARCHAR(30),
->     CGPA DECIMAL(3,2)
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO STUDENT VALUES
-> (101,'Rahul','CSE',8.5),
-> (102,'Priya','IT',9.1),
-> (103,'Arun','ECE',7.8),
-> (104,'Meena','AIDS',8.9),
-> (105,'Kiran','CSE',8.2);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM STUDENT;
+-----+-----+-----+-----+
| SID | Name | Branch | CGPA |
+-----+-----+-----+-----+
| 101 | Rahul | CSE    | 8.50 |
| 102 | Priya | IT     | 9.10 |
| 103 | Arun  | ECE    | 7.80 |
| 104 | Meena | AIDS   | 8.90 |
| 105 | Kiran | CSE    | 8.20 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT *
-> FROM STUDENT
-> WHERE CGPA>8.0
-> ORDER BY CGPA DESC;
+-----+-----+-----+-----+
| SID | Name | Branch | CGPA |
+-----+-----+-----+-----+
| 102 | Priya | IT     | 9.10 |
| 104 | Meena | AIDS   | 8.90 |
| 101 | Rahul | CSE    | 8.50 |
| 105 | Kiran | CSE    | 8.20 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Question 3:For tables ORDERS(OID, CustID, Date, Amount) and CUSTOMER(CustID, Name, City), write an SQL JOIN query to find all orders placed by customers from Chennai.

Aim:To retrieve all orders placed by customers from Chennai using an SQL JOIN.

Query

```
SELECT O.OID, O.CustID, C.Name, C.City, O.Date, O.Amount
```

FROM ORDERS O

JOIN CUSTOMER C

ON O.CustID = C.CustID

WHERE C.City = 'Chennai';

```
MySQL 8.0 Command Line Cli x + v
-> (101,'Rahul','Chennai'),
-> (102,'Priya','Bangalore'),
-> (103,'Arun','Chennai');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO ORDERS VALUES
-> (1,101,'2024-01-10',5000),
-> (2,102,'2024-01-15',3000),
-> (3,103,'2024-01-20',4500);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM CUSTOMER;
+-----+-----+-----+
| CustID | Name  | City    |
+-----+-----+-----+
| 101    | Rahul | Chennai |
| 102    | Priya | Bangalore |
| 103    | Arun  | Chennai |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM ORDERS;
+-----+-----+-----+-----+
| OID | CustID | Date       | Amount |
+-----+-----+-----+-----+
| 1   | 101    | 2024-01-10 | 5000.00 |
| 2   | 102    | 2024-01-15 | 3000.00 |
| 3   | 103    | 2024-01-20 | 4500.00 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT O.OID,O.CustID,C.Name,C.City,O.Date,O.Amount
-> FROM ORDERS O
-> JOIN CUSTOMER C
-> ON O.CustID=C.CustID
-> WHERE C.City='Chennai';
+-----+-----+-----+-----+-----+-----+
| OID | CustID | Name  | City    | Date       | Amount |
+-----+-----+-----+-----+-----+-----+
| 1   | 101    | Rahul | Chennai | 2024-01-10 | 5000.00 |
| 3   | 103    | Arun  | Chennai | 2024-01-20 | 4500.00 |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Question5: Using relational algebra, perform a natural join between DEPT(DeptNo, DeptName) and EMP(EmpNo, EmpName, DeptNo) and project only EmpName and DeptName.

Aim: To perform a natural join between DEPT and EMP tables and display only the employee name and department name.

Relational Algebra Query

$$\pi_{EmpName, DeptName}(EMP \bowtie DEPT)$$

```
MySQL 8.0 Command Line Cli x + v

mysql> INSERT INTO DEPT VALUES
  -> (10,'HR'),
  -> (20,'IT'),
  -> (30,'Finance');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO EMP VALUES
  -> (101,'Rahul',10),
  -> (102,'Priya',20),
  -> (103,'Arun',30);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM DEPT;
+-----+-----+
| DeptNo | DeptName |
+-----+-----+
|      10 | HR       |
|      20 | IT       |
|      30 | Finance  |
+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM EMP;
+-----+-----+-----+
| EmpNo | EmpName | DeptNo |
+-----+-----+-----+
|    101 | Rahul   |      10 |
|    102 | Priya   |      20 |
|    103 | Arun    |      30 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT E.EmpName,D.DeptName
  -> FROM EMP E
  -> NATURAL JOIN DEPT D;
+-----+-----+
| EmpName | DeptName |
+-----+-----+
| Rahul   | HR       |
| Priya   | IT       |
| Arun    | Finance  |
+-----+-----+
3 rows in set (0.00 sec)
```

Question 5: Write an SQL subquery to find the employee who earns the maximum salary in each department from EMPLOYEE(EmpID, Name, Dept, Salary).

Aim : To find the employee who earns the highest salary in each department using an SQL subquery.

Query

```
SELECT EmpID, Name, Dept, Salary
FROM EMPLOYEE E
WHERE Salary =
(
    SELECT MAX(Salary)
    FROM EMPLOYEE
    WHERE Dept = E.Dept
);
```



```
MySQL 8.0 Command Line Cli  X + v
mysql> CREATE TABLE EMPLOYEE
-> (
->     EmpID INT PRIMARY KEY,
->     Name VARCHAR(50),
->     Dept VARCHAR(30),
->     Salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO EMPLOYEE VALUES
-> (101,'Rahul','HR',60000),
-> (102,'Priya','IT',75000),
-> (103,'Arun','HR',55000),
-> (104,'Meena','Finance',70000),
-> (105,'Kiran','IT',80000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM EMPLOYEE;
+-----+-----+-----+-----+
| EmpID | Name  | Dept  | Salary |
+-----+-----+-----+-----+
| 101   | Rahul | HR    | 60000.00 |
| 102   | Priya | IT    | 75000.00 |
| 103   | Arun  | HR    | 55000.00 |
| 104   | Meena | Finance | 70000.00 |
| 105   | Kiran | IT    | 80000.00 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> SELECT EmpID,Name,Dept,Salary
-> FROM EMPLOYEE E
-> WHERE Salary=
-> (
-> SELECT MAX(Salary)
-> FROM EMPLOYEE
-> WHERE Dept=E.Dept
-> );
+-----+-----+-----+-----+
| EmpID | Name  | Dept  | Salary |
+-----+-----+-----+-----+
| 101   | Rahul | HR    | 60000.00 |
| 104   | Meena | Finance | 70000.00 |
| 105   | Kiran | IT    | 80000.00 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```